

PART 1 — SECTION 1 (Q1–20)

Amorphous & Crystalline Solids — Detailed Answers

1. Define crystalline solids.

Crystalline solids are solids in which the constituent particles (atoms, ions, or molecules) are arranged in a **definite, repeating three-dimensional pattern** known as a crystal lattice. They possess **long-range order**, meaning their arrangement repeats over large distances, resulting in sharp melting points and anisotropic properties.

2. Define amorphous solids.

Amorphous solids are solids whose constituent particles do **not** have a regular, repeating arrangement. They possess only **short-range order** and behave more like extremely viscous liquids. They do not have sharp melting points and soften over a range of temperatures.

3. Give one example each of crystalline and amorphous solids.

- **Crystalline solid:** Sodium chloride (NaCl), quartz, copper.
 - **Amorphous solid:** Glass, rubber, plastics.
-

4. Difference between crystalline and amorphous solids in terms of long-range order.

- **Crystalline solids:** Exhibit **long-range order**, where particle arrangement repeats over long distances.
 - **Amorphous solids:** Lack long-range order and show only **short-range order** limited to nearest neighbours.
-

5. What is isotropy? Which solids show isotropic properties?

Isotropy is a property where a material exhibits **same physical properties** (refractive index, thermal expansion, electrical conductivity) in **all directions**.

Amorphous solids are isotropic because their particles are arranged randomly in all directions.

6. Define anisotropy. Give example.

Anisotropy is the property due to which a material shows **different physical properties in different directions**.

Example: Crystalline solids such as calcite and quartz show anisotropy because the arrangement of particles varies with direction.

7. Explain how melting points differ for crystalline and amorphous solids.

- **Crystalline solids:** Have **sharp melting points** because their particles are arranged uniformly; the entire structure breaks at one specific temperature.
 - **Amorphous solids:** Do **not** have a sharp melting point; instead, they **soften gradually** over a range of temperatures due to non-uniform arrangement.
-

8. Give examples of amorphous solids used in everyday life.

- Glass (windows, bottles)
 - Plastics (polythene bags, PVC pipes)
 - Rubber (tyres, seals)
 - Gels, wax, pitch
 - Fiberglass
-

9. Define polycrystalline solids.

Polycrystalline solids consist of a very large number of **small crystals or grains**, each with its own orientation. The boundaries between these grains are called **grain boundaries**.

Metals like iron and copper are polycrystalline.

10. Explain how glasses are formed.

Glasses are formed when certain materials (like silica) are **cooled rapidly (quenched)** from their molten state. The rapid cooling prevents the particles from arranging into a crystalline lattice, resulting in a **supercooled liquid structure**, i.e., an amorphous solid.

11. Define allotropy. Give example.

Allotropy is the phenomenon in which an element exists in **two or more different structural forms** in the same physical state.

Example: Carbon exists as diamond, graphite, graphene, and fullerenes.

12. Difference between single crystal and polycrystalline solids.

- **Single crystal:** Entire solid consists of **one continuous crystal**, same orientation throughout (e.g., silicon wafers).
 - **Polycrystalline:** Made of many small crystals (grains) with different orientations (e.g., metals like iron, copper).
-

13. Which solids show sharp melting points?

Crystalline solids show sharp melting points due to their ordered, uniform arrangement.

14. What is short-range order? Which type of solid exhibits it?

Short-range order means that the arrangement of particles is regular **only up to a few atomic distances**, beyond which the order is lost.

Amorphous solids exhibit short-range order.

15. Define unit cell in crystalline solids.

A unit cell is the **smallest repeating structural unit** of a crystal lattice that, when repeated in all directions, generates the entire crystal structure. It defines the lattice parameters and symmetry.

16. Give examples of crystalline solids with metallic bonding.

- Iron (Fe)
- Copper (Cu)
- Silver (Ag)
- Gold (Au)
- Sodium (Na)

These consist of positive metal ions in a sea of delocalized electrons.

17. Give examples of crystalline solids with ionic bonding.

- Sodium chloride (NaCl)
 - Potassium bromide (KBr)
 - Magnesium oxide (MgO)
 - Calcium fluoride (CaF₂)
-

18. Give examples of crystalline solids with covalent bonding.

- Diamond (3D network)
 - Quartz (SiO₂)
 - Graphite
 - Silicon carbide (SiC)
-

19. Explain molecular solids with example.

Molecular solids are solids in which particles are **molecules** held together by **weak intermolecular forces** (Van der Waals forces, dipole interactions, hydrogen bonds).

Example: Ice (H₂O), solid CO₂ (dry ice), iodine crystals.

20. What is the coordination number in ionic solids?

The coordination number is the **number of oppositely charged ions surrounding a given ion** in the crystal structure.

Example values:

- NaCl : 6:6
- CsCl : 8:8

- ZnS (zinc blende) : 4:4
-

SECTION 1 Completed (Q1–20).

Would you like me to continue with

SECTION 2 (Q21–50): Crystal Lattices & Unit Cells?

yes

Great!

Here are full, detailed, exam-oriented answers for SECTION 2 (Q21–50).

SECTION 2: Crystal Lattices & Unit Cells (Q21–50)

Detailed Answers (Option A)

21. Define crystal lattice.

A crystal lattice is a **three-dimensional arrangement of points** in space representing the positions of atoms, ions, or molecules in a crystalline solid. Each point in the lattice has an identical environment, and the entire crystal structure can be generated by repeating the lattice in all directions.

22. Define lattice points.

Lattice points are the **positions in a crystal lattice** where atoms, ions, or molecules are assumed to be located. Each lattice point represents one or more particles and carries complete information about the environment of the crystal structure.

23. Differentiate between lattice and motif.

- **Lattice:** A regular, periodic arrangement of points in space.

- **Motif (basis):** A group of atoms associated with each lattice point.

✚ Crystal structure = Lattice + Motif

24. What is primitive unit cell?

A primitive unit cell (simple unit cell) is a unit cell that contains **only one lattice point** per cell. Atoms are present **only at the corners**, and each corner atom contributes only 1/8th to the cell.

25. What is body-centered cubic (BCC) unit cell?

A BCC unit cell consists of:

- 8 atoms at corners, and
- 1 atom at the **center of the cube**.

Total atoms = 2 per unit cell.

The body-centered atom does not share space—belongs entirely to that cell.

26. What is face-centered cubic (FCC) unit cell?

An FCC unit cell has:

- 8 corner atoms
- 6 atoms at the **centers of faces**

Total atoms = 4 per unit cell.

27. Give examples of BCC metals.

- Iron (Fe) at room temperature (α -iron)
 - Chromium (Cr)
 - Tungsten (W)
 - Sodium (Na)
 - Potassium (K)
 - Molybdenum (Mo)
-

28. Give examples of FCC metals.

- Aluminium (Al)
 - Copper (Cu)
 - Gold (Au)
 - Silver (Ag)
 - Lead (Pb)
 - Nickel (Ni)
-

29. Define simple cubic (SC) unit cell.

An SC unit cell contains atoms **only at the 8 corners** of the cube.

Total atoms = 1 per unit cell (8 corner atoms $\times$ 1/8 contribution).

30. Number of atoms in SC, BCC, FCC unit cells.

- SC: 1 atom
 - BCC: 2 atoms
 - FCC: 4 atoms
-

31. Edge length in terms of atomic radius for SC.

In SC, atoms touch along the edge.

Thus,

$$a = 2r$$

(a = edge length, r = atomic radius)

32. Edge length in terms of atomic radius for BCC.

In BCC, atoms touch along the **body diagonal**, whose length is $4r$.

$$\sqrt{3}a = 4r$$

$$a = \frac{4r}{\sqrt{3}}$$

33. Edge length in terms of atomic radius for FCC.

Atoms touch along the **face diagonal**, whose length is $4r$.

$$\sqrt{2}a = 4r$$

$$a = \frac{4r}{\sqrt{2}} = 2\sqrt{2}r$$

34. Define packing efficiency.

Packing efficiency is the **percentage of total space occupied by atoms** in a unit cell.

$$\text{Packing Efficiency} = \frac{\text{Volume occupied by atoms}}{\text{Volume of unit cell}} \times 100$$

35. Packing efficiency for SC, BCC, FCC, HCP.

- SC: 52.4 %
- BCC: 68 %
- FCC: 74 %
- HCP: 74 %

(FCC and HCP are most efficient packings.)

36. Difference between primitive and non-primitive unit cells.

- **Primitive (P):** Atoms only at corners; 1 atom per cell.
 - **Non-primitive:** Have additional atoms at body center (BCC), face centers (FCC), or edges.
-

37. Define close-packed structure.

A close-packed structure is the arrangement of atoms in which **maximum space is filled** and voids are minimized. It includes:

- HCP (hexagonal close packing)
- CCP/FCC (cubic close packing)

Both achieve 74% packing efficiency.

38. Examples of metals with HCP structure.

- Magnesium (Mg)
 - Zinc (Zn)
 - Titanium (Ti)
 - Cobalt (Co)
 - Beryllium (Be)
-

39. Relationship between density, atomic mass, and unit cell dimensions.

$$\rho = \frac{Z \times M}{N_A \times a^3}$$

Where:

- Z = no. of atoms in unit cell
- M = molar mass
- a = edge length
- N_A = Avogadro number

Used to calculate density of crystalline solids.

40. Concept of lattice constant.

The lattice constant is the **edge length (a)** of a unit cell, or the **distance between repeating points** in a lattice. For cubic systems, it is simply the cube edge; for tetragonal, orthorhombic systems, a, b, c differ.

41. Number of atoms in SC unit cell.

SC has 8 corner atoms, each contributing 1/8.

$$8 \times \frac{1}{8} = 1 \text{ atom}$$

42. Number of atoms in BCC unit cell.

8 corner atoms + 1 body atom

$$8 \times \frac{1}{8} + 1 = 2 \text{ atoms}$$

43. Number of atoms in FCC unit cell.

8 corner atoms + 6 face atoms

$$8 \times \frac{1}{8} + 6 \times \frac{1}{2} = 4 \text{ atoms}$$

44. Number of atoms in HCP unit cell.

HCP contains **6 atoms** per unit cell.

(12 corner atoms $\times$ 1/6 + 2 center atoms = 6)

45. Derive number of atoms in SC unit cell.

- 8 corners $\times$ each 1/8 contribution

$$N = 8 \times \frac{1}{8} = 1$$

Thus, SC contains **1 atom**.

46. Derive number of atoms in BCC unit cell.

- 8 corner atoms: $8 \times 1/8 = 1$
- 1 body atom: full contribution

Total = **2 atoms**

47. Derive number of atoms in FCC unit cell.

- Corners: $8 \times 1/8 = 1$
- Faces: $6 \times 1/2 = 3$

Total = **4 atoms**

48. Derive number of atoms in HCP unit cell.

- 12 corner atoms $\times$ 1/6 = 2

- 3 atoms in middle layer (full)
 - 1 atom at center
- Total = 6 atoms per HCP unit cell
-

49. Fractional contribution of corner atom in a unit cell.

A corner atom is shared by 8 neighbouring unit cells.

Contribution =

$$\frac{1}{8}$$

50. Fractional contribution of face-centered atom.

A face atom is shared by 2 unit cells.

Contribution =

$$\frac{1}{2}$$

If you'd like, I can continue immediately with the next part (Q51–70).

Shall I proceed to SECTION 3: Number of Atoms in Unit Cell (Q51–70)?

yes

Perfect!

Here are full, detailed, exam-oriented answers for SECTION 3 (Q51–70).

SECTION 3: Number of Atoms in Unit Cell (Q51–70)

Detailed Answers (Option A)

51. Fractional contribution of body-centered atom.

A body-centered atom is located **wholly inside** the unit cell, so it is **not shared** with any other cell.

Thus contribution:

$$1$$

52. Fractional contribution of edge atom.

An edge atom is shared by **4 unit cells** because each edge is common to four adjacent cubes.

Thus, contribution:

$$\frac{1}{4}$$

53. How many lattice points in simple cubic (SC) unit cell?

In SC, lattice points exist **only at the corners**.

- 8 corners $\times$ 1 lattice point each = **8 lattice points**

Although only 1 atom per unit cell, the number of lattice **points** = 8.

54. How many lattice points in BCC unit cell?

BCC has:

- 8 lattice points at corners
- 1 lattice point at body center

Total lattice points = **9**

55. How many lattice points in FCC unit cell?

FCC has:

- 8 lattice points at corners
- 6 lattice points at face centers

Total = **14 lattice points**

56. How many lattice points in HCP unit cell?

HCP has:

- 12 corner points
- 2 face-centered points (top & bottom faces)
- 3 points inside the cell

Total = 17 lattice points

57. Calculate number of atoms per unit volume for FCC metal.

Number of atoms per unit cell = 4

Volume of cell = a^3

Thus,

$$\text{Atoms per unit volume} = \frac{4}{a^3}$$

If asked in terms of atomic radius r :

$$a = 2\sqrt{2}r$$

$$\text{Atoms per unit volume} = \frac{4}{(2\sqrt{2}r)^3}$$

58. Calculate number of atoms per unit volume for BCC metal.

Atoms per unit cell = 2

$$\text{Atoms per unit volume} = \frac{2}{a^3}$$

If written using radius:

$$a = \frac{4r}{\sqrt{3}}$$

$$\text{Atoms per unit volume} = \frac{2}{\left(\frac{4r}{\sqrt{3}}\right)^3}$$

59. Differentiate atomic packing factor (APF) and packing efficiency.

Atomic packing factor (APF):

Fraction of unit cell volume actually occupied by atoms

$$APF = \frac{\text{Volume of atoms}}{\text{Volume of unit cell}}$$

Packing efficiency:

Percentage of space occupied

$$\text{Packing efficiency} = APF \times 100$$

✚ APF is a **fraction**; Packing efficiency is a **percentage**.

60. Examples of metals with SC structure.

Simple cubic metals are rare. The only common example is:

- Polonium (Po)

Most metals prefer BCC, FCC, or HCP due to higher packing efficiency.

SECTION 3 Completed!

Next up is:

SECTION 4 (Q71–100): Packing Efficiency & Voids

Shall I continue?

yes

Excellent!

Here are **full, detailed, exam-oriented** answers for SECTION 4 (Q71–100).

SECTION 4: Packing Efficiency & Voids (Q71–100)

Detailed Answers (Option A)

61. Define packing efficiency.

Packing efficiency is the **percentage of total space occupied by atoms** in a unit cell.

$$\text{Packing efficiency} = \frac{\text{Volume of atoms in the cell}}{\text{Total volume of unit cell}} \times 100$$

It indicates how tightly atoms are packed in a crystal.

62. Packing efficiency of SC.

In a simple cubic cell:

- Atoms touch along the edge $\rightarrow a = 2r$
- Volume of atoms = volume of 1 sphere

$$= \frac{4}{3}\pi r^3$$

- Volume of cell $= (2r)^3 = 8r^3$

Packing efficiency:

$$= \frac{\frac{4}{3}\pi r^3}{8r^3} \times 100 = 52.4\%$$

63. Packing efficiency of BCC.

- Atoms touch along body diagonal $\rightarrow \sqrt{3}a = 4r$
- 2 atoms per cell

Total volume of atoms:

$$= 2 \times \frac{4}{3}\pi r^3$$

Volume of unit cell:

$$a^3 = \left(\frac{4r}{\sqrt{3}}\right)^3$$

Final result:

$$= 68\%$$

64. Packing efficiency of FCC.

In FCC, atoms touch along face diagonal.

- 4 atoms per cell
- $APF = 0.74$

Thus, packing efficiency =

74%

65. Packing efficiency of HCP.

HCP has the same close-packed arrangement as FCC.

Thus, packing efficiency =

74%

66. Why SC has lowest packing efficiency?

SC has atoms only at the corners, leaving large empty spaces in the center and along the edges.

Since atoms touch only along **one direction (edge)**, the packing is loose, resulting in very low packing efficiency (~52.4%).

67. Define void in crystal structure.

A void is an **empty space** between closely packed atoms where no atom is present. These are created because spheres cannot fill all space completely in 3D packing.

68. Voids in SC, BCC, FCC, HCP.

SC:

- Very large voids
- No standard tetrahedral/octrahedral voids because atoms are not close-packed

BCC:

- Irregular voids
- No regular tetrahedral voids
- Some distorted octahedral voids

FCC:

- 8 tetrahedral voids
- 4 octahedral voids per unit cell

HCP:

- Same as FCC (close-packed)
 - 2 octahedral voids
 - 4 tetrahedral voids (per primitive cell)
-

69. Tetrahedral voids: definition and examples.

A tetrahedral void is formed when **four atoms** are arranged at the corners of a tetrahedron, leaving a small space in the center.

Examples:

- Zn^{2+} in ZnS (zinc blende) occupies tetrahedral voids
 - Li^+ in Li_2O
-

70. Octahedral voids: definition and examples.

An octahedral void is a space surrounded by **six atoms** arranged octahedrally (like eight-pointed shape).

Examples:

- Na^+ occupies octahedral voids in NaCl
 - Mg^{2+} in MgO
 - Al^{3+} in Al_2O_3
-

71. Fraction of tetrahedral voids per atom in FCC.

FCC has:

- 4 atoms per cell
- 8 tetrahedral voids

Thus,

$$\text{Void/atom} = \frac{8}{4} = 2$$

2 tetrahedral voids per atom

72. Fraction of octahedral voids per atom in FCC.

FCC has:

- 4 atoms
- 4 octahedral voids

$$\frac{4}{4} = 1$$

1 octahedral void per atom

73. Fraction of tetrahedral voids per atom in BCC.

BCC is **not close-packed**, so the tetrahedral voids are irregular.

Approx value:

$$\text{Tetrahedral voids per atom} \approx 6$$

(but not used in most exam calculations)

74. Fraction of octahedral voids per atom in BCC.

BCC has approx 3 octahedral voids per atom, but again these are **distorted** and not used often.

75. Explain why HCP and FCC have same packing efficiency.

Both FCC and HCP are **close-packed structures** where atoms occupy optimum space.

- Both follow **ABCABC (FCC)** or **ABAB (HCP)** stacking
- Both have same coordination number (12)

Thus, both achieve **74% packing efficiency**.

76. Relation between radius of atom and size of tetrahedral void.

Let:

- r = radius of atom

- R = radius of tetrahedral void

Formula:

$$R = 0.225r$$

77. Relation between radius of atom and size of octahedral void.

Let R = radius of void

$$R = 0.414r$$

78. Calculate radius of tetrahedral void in terms of atomic radius.

$$R_{tet} = 0.225r$$

79. Calculate radius of octahedral void in terms of atomic radius.

$$R_{oct} = 0.414r$$

80. Difference between interstitial site and void.

Void:

- Naturally occurring empty space in packing
- Formed due to geometry of spheres

Interstitial site:

- A void that is **occupied by a smaller atom or ion**
 - Example: carbon occupies interstitial sites in iron (steel)
-

SECTION 4 Completed!

Next is:

SECTION 5 (Q101–120): Density of Unit Cell

Would you like me to continue?

yes

Perfect!

Here are full, detailed, exam-oriented answers for SECTION 5 (Q101–120).

SECTION 5: Density of Unit Cell (Q101–120)

Detailed Answers (Option A)

81. Derive formula for density of unit cell.

Density (ρ) is mass per unit volume.

- Mass of unit cell = number of atoms $\times$ mass of one atom

$$m = Z \times \frac{M}{N_A}$$

Where:

- Z = number of atoms per unit cell
- M = molar mass
- N_A = Avogadro number
- Volume of unit cell = a^3 (for cubic cells)

$$\therefore \rho = \frac{\text{Mass}}{\text{Volume}} = \frac{ZM}{N_A a^3}$$

82. Density in terms of atomic mass, number of atoms, and volume.

$$\rho = \frac{Z \cdot M}{N_A \cdot V_{\text{cell}}}$$

Where V_{cell} is the volume of the unit cell.

83. Calculate density of SC metal with given atomic radius and mass.

- For SC: $a = 2r$, atoms per cell $Z = 1$

- Mass per unit cell = M/N_A
- Volume = $(2r)^3$

$$\rho = \frac{1 \cdot M}{N_A \cdot (2r)^3}$$

84. Calculate density of BCC metal.

- BCC: $a = 4r/\sqrt{3}$, atoms per cell $Z = 2$

$$\rho = \frac{2 \cdot M}{N_A \cdot (4r/\sqrt{3})^3}$$

85. Calculate density of FCC metal.

- FCC: $a = 2\sqrt{2}r$, atoms per cell $Z = 4$

$$\rho = \frac{4 \cdot M}{N_A \cdot (2\sqrt{2}r)^3}$$

86. Explain why FCC metals are denser than SC metals.

- FCC has more atoms per unit cell (4 vs 1)
 - FCC is close-packed → smaller void fraction
 - Hence, mass/volume (density) is higher in FCC than SC.
-

87. Relation between atomic packing factor and density.

- Higher **APF** → more atoms occupy unit cell → higher mass per unit volume → higher **density**
 - $\rho \propto APF$
-

88. Calculate density for HCP structure.

- Atoms per HCP cell $Z = 6$
- Unit cell volume: $V = \frac{3\sqrt{3}}{2}a^2c$ (a = base edge, c = height)

$$\rho = \frac{6 \cdot M}{N_A \cdot V}$$

89. Concept of molar volume in solids.

Molar volume (V_m) = volume occupied by 1 mole of solid

$$V_m = \frac{M}{\rho}$$

Where M = molar mass, ρ = density.

90. Difference between theoretical and experimental density.

- **Theoretical density:** Calculated from unit cell dimensions, atomic mass, and crystal structure
 - **Experimental density:** Measured using methods like **pycnometer** or **floatation**
 - Difference arises due to defects, porosity, or impurities.
-

91. Effect of voids on density of solid.

- More voids → less mass per unit volume → lower density
 - Close-packed structures (FCC, HCP) → fewer voids → higher density
-

92. Density calculation of ionic solids using unit cell.

For ionic solids:

$$\rho = \frac{\text{mass of formula units in cell}}{\text{volume of unit cell}}$$

- Mass = $Z \times \text{molar mass of AB} / N_A$
- Volume = a^3

Example: NaCl (FCC structure, $Z = 4$)

93. Calculate lattice parameter from density and atomic mass.

$$a = \left(\frac{Z \cdot M}{N_A \cdot \rho} \right)^{1/3}$$

- Z = atoms per cell

- M = molar mass
 - ρ = density
 - Gives cubic lattice edge
-

94. Convert atomic radius to edge length.

Depends on crystal structure:

- SC: $a = 2r$
 - BCC: $a = \frac{4r}{\sqrt{3}}$
 - FCC: $a = 2\sqrt{2}r$
 - HCP: a = base edge; $c = 1.633a$ (ideal)
-

95. Calculate number of formula units in unit cell from density.

$$Z = \frac{\rho \cdot N_A \cdot V_{cell}}{M}$$

Where:

- ρ = density
 - V_{cell} = unit cell volume
 - M = molar mass
-

96. Calculate atomic radius from density of FCC metal.

1. Density formula: $\rho = \frac{4M}{N_A a^3}$
 2. FCC: $a = 2\sqrt{2}r \rightarrow r = \frac{a}{2\sqrt{2}}$
 3. Substitute measured density $\rightarrow$ calculate r
-

97. Calculate atomic radius from density of BCC metal.

1. Density formula: $\rho = \frac{2M}{N_A a^3}$
 2. BCC: $a = 4r/\sqrt{3} \rightarrow r = \frac{\sqrt{3}a}{4}$
 3. Substitute density $\rightarrow$ solve for r
-

98. Compare densities of FCC and BCC metals of same atomic mass.

- FCC: 4 atoms/unit cell, packing efficiency = 74%
- BCC: 2 atoms/unit cell, packing efficiency = 68%

FCC metals are denser because more atoms occupy the same volume.

99. Density of compound AB in FCC structure.

- $Z = 4$ (FCC)
- Mass of AB in unit cell = $4 \times (\text{mass of AB}) = 4 \cdot M_{AB}/N_A$
- Volume = a^3

$$\rho = \frac{4 \cdot M_{AB}}{N_A \cdot a^3}$$

100. Density of compound AB in BCC structure.

- $Z = 2$ (BCC)
- Mass = $2 \times M/N_A$
- Volume = a^3

$$\rho = \frac{2 \cdot M_{AB}}{N_A \cdot a^3}$$

SECTION 5 Completed!

Next up is:

SECTION 6 (Q121–140): Imperfections in Solids

Shall I continue?

yes make it till 150

Perfect! Let's complete all remaining questions (Q121–150) covering Imperfections, Electrical & Magnetic Properties with detailed answers.

SECTION 6: Imperfections in Solids (Q121–140)

101. Define point defects in solids.

Point defects are **localized disruptions** in the crystal lattice involving **one or few atoms**. Examples include vacancies, interstitials, and substitutional defects.

102. Examples of point defects.

- **Vacancy** – missing atom (e.g., NaCl Schottky defect)
 - **Interstitial** – atom in an unusual site
 - **Substitutional impurity** – foreign atom replacing host atom
-

103. Schottky defect: definition and example.

- **Definition:** Equal number of cations and anions are missing from the lattice, maintaining charge neutrality.
- **Example:** NaCl, KCl, CsCl

Effect: Decreases density because atoms are missing.

104. Frenkel defect: definition and example.

- **Definition:** Cation leaves its normal position and occupies an interstitial site.
- **Example:** AgCl, ZnS

Effect: Density remains almost unchanged.

105. Effect of Schottky defect on density.

- Density **decreases** due to missing atoms.
-

106. Effect of Frenkel defect on density.

- Density remains **nearly unchanged** because atom is just displaced, not removed.
-

107. Line defects: definition.

Line defects (dislocations) are **one-dimensional defects** in which atoms are misaligned along a line in the crystal.

108. Edge dislocation: definition.

- Extra half-plane of atoms inserted into a crystal
 - Motion of dislocations allows **plastic deformation**
-

109. Screw dislocation: definition.

- Spiral planar ramp around a line defect
 - Atoms are displaced along the axis of the dislocation
-

110. Effects of dislocations on mechanical properties.

- Increase plasticity
 - Cause strain hardening
 - Affect yield strength and ductility
-

111. Surface defects: definition.

- Imperfections along surfaces or grain boundaries, such as steps, ledges, and terraces.
-

112. Grain boundaries: definition.

- Interface between two differently oriented crystals (grains) in polycrystalline materials.
-

113. Stacking faults: definition.

- Error in the regular stacking sequence of close-packed planes (ABCABC → ABCBC).
-

114. Colour centres: definition and examples.

- Defects that **absorb light** and give color to ionic crystals.
 - **Examples:** F-centres in NaCl (electron in a vacancy)
-

115. Defects in ionic solids: examples.

- Schottky defect: NaCl
 - Frenkel defect: AgCl, ZnS
-

116. Intrinsic vs extrinsic defects.

- **Intrinsic:** Due to lattice atoms themselves (e.g., Schottky, Frenkel)
 - **Extrinsic:** Due to impurities or foreign atoms
-

117. Imperfections in metals.

- Vacancies
 - Dislocations (edge, screw)
 - Substitutional or interstitial impurities
-

118. Imperfections in ceramics.

- Schottky and Frenkel defects
 - Grain boundaries
 - Porosity
-

119. Imperfections in semiconductors.

- Vacancies, interstitials
 - Dopant atoms (n-type, p-type)
 - Dislocations affecting carrier mobility
-

120. Relationship between defect concentration and temperature.

- Defect concentration increases **exponentially with temperature**

$$n = N \exp\left(-\frac{Q}{kT}\right)$$

Where Q = defect formation energy, k = Boltzmann constant

SECTION 7: Electrical & Magnetic Properties (Q141–150)

121. Define electrical conductivity in solids.

- Electrical conductivity (σ) is the ability of a solid to **conduct electric current**.

$$\sigma = \frac{1}{\rho} \quad (\rho = \text{resistivity})$$

122. Examples of good conductors.

- Copper (Cu)
 - Silver (Ag)
 - Gold (Au)
-

123. Examples of insulators.

- Glass
 - Rubber
 - Mica
-

124. Examples of semiconductors.

- Silicon (Si)
 - Germanium (Ge)
-

125. Define intrinsic semiconductors.

- Pure semiconductor with no impurities
- Conductivity arises from **electron-hole pairs** generated thermally

126. Define extrinsic semiconductors.

- Semiconductor **doped with impurities** to increase conductivity
 - Example: n-type (phosphorus in Si), p-type (boron in Si)
-

127. Difference between n-type and p-type semiconductors.

Property	n-type	p-type
Dopant	Pentavalent	Trivalent
Charge carriers	Electrons	Holes
Example	P in Si	B in Si

128. Factors affecting electrical conductivity.

- Temperature ($\uparrow T \rightarrow \uparrow$ conductivity for semiconductors)
 - Impurities
 - Defects and crystal structure
 - Carrier concentration
-

129. Define magnetic susceptibility.

- Measure of how much a material **becomes magnetized** in an applied magnetic field.

$$\chi = \frac{M}{H} \quad (M = \text{magnetization}, H = \text{magnetic field})$$

130. Types of magnetism: diamagnetism, paramagnetism, ferromagnetism.

- **Diamagnetism:** Weakly repelled by magnetic field (all electrons paired)
 - **Paramagnetism:** Weakly attracted (unpaired electrons)
 - **Ferromagnetism:** Strongly attracted, retains magnetization (magnetic domains)
-

131. Examples of diamagnetic materials.

- Bismuth (Bi)
 - Copper (Cu)
 - Silver (Ag)
-

132. Examples of paramagnetic materials.

- Aluminium (Al)
 - Platinum (Pt)
 - Oxygen (O₂)
-

133. Examples of ferromagnetic materials.

- Iron (Fe)
 - Nickel (Ni)
 - Cobalt (Co)
-

134. Effect of temperature on magnetic properties.

- Paramagnetism: decreases with temperature (Curie-Weiss law)
 - Ferromagnetism: disappears above **Curie temperature**
-

135. Curie temperature definition.

- Temperature above which a ferromagnetic material becomes **paramagnetic**.
-

136. Magnetic domains: definition.

- Small regions in a ferromagnet where magnetic moments are **aligned in the same direction**
-

137. Magnetic hysteresis: definition.

- Lag between magnetization (M) and applied field (H)
- Shown as a **hysteresis loop**

138. Applications of magnetic materials.

- Transformers
 - Motors
 - Magnetic storage devices
-

139. Superconductors: definition and examples.

- Materials with **zero electrical resistance** below critical temperature T_c
 - Examples: Hg, Pb, $\text{YBa}_2\text{Cu}_3\text{O}_7$
-

140. Critical temperature of superconductor.

- Temperature below which superconductivity appears (resistance = 0)
-

141. Meissner effect: definition.

- Complete **expulsion of magnetic field** from a superconductor below T_c
-

142. Applications of superconductors in MRI and maglev.

- MRI: High magnetic fields for imaging
 - Maglev trains: Magnetic levitation reduces friction
-

143. Electrical conductivity of metals vs semiconductors.

- Metals: High conductivity, decreases with temperature
 - Semiconductors: Low conductivity, increases with temperature
-

144. Hall effect: definition and applications.

- **Voltage developed perpendicular** to current and magnetic field in a conductor
- Applications:
 - Determine carrier type (n or p)
 - Measure magnetic field

145. Nernst effect: definition.

- Generation of **transverse electric field** when a temperature gradient is applied in presence of a magnetic field

146. Piezoelectric solids: definition.

- Solids that **develop electric potential** when mechanically stressed

147. Examples of piezoelectric solids.

- Quartz (SiO_2)
- Rochelle salt
- Barium titanate (BaTiO_3)

148. Relationship between crystal structure and electrical properties.

- Symmetry and atom arrangement affect **band structure**, defects, and carrier mobility
→ determine conductivity, semiconductivity, or insulation

149. Effect of defects on conductivity.

- Vacancies and interstitials can **increase/decrease** carrier concentration
- Dislocations scatter electrons → reduce mobility

150. Effect of impurities on magnetic properties.

- Can enhance or reduce magnetization
- Example: adding non-magnetic impurities reduces ferromagnetism
- Doping with magnetic ions can increase magnetic ordering